

LINUX CMDS

1. Ping : Ping is a network troubleshooting tool used to test reachability between two devices. It measures response time and packet loss.

Protocols Involved in Ping

ICMP (Internet Control Message Protocol)

Ping mainly uses ICMP. It sends an Echo Request to the destination and waits for an Echo Reply. This helps verify whether the remote device is reachable.

ARP (Address Resolution Protocol)

Before sending packets inside a local network, the device must know the destination MAC address. ARP resolves the IP address into a MAC address.

IP Protocol (Layer 3)

IP handles logical addressing and routing. The packet carries source and destination IP addresses so routers know where to forward it.

DNS Eg: If you ping a domain name like [google.com](https://www.google.com), DNS first translates the domain into an IP address.

Step-by-Step Flow

User types: "ping [google.com](https://www.google.com)"

DNS resolves the domain into an IP address

ARP finds the MAC address (inside LAN)

ICMP Echo Request is sent

Destination replies with ICMP Echo Reply

You see time, TTL, and packet statistics

Why Ping is Important?

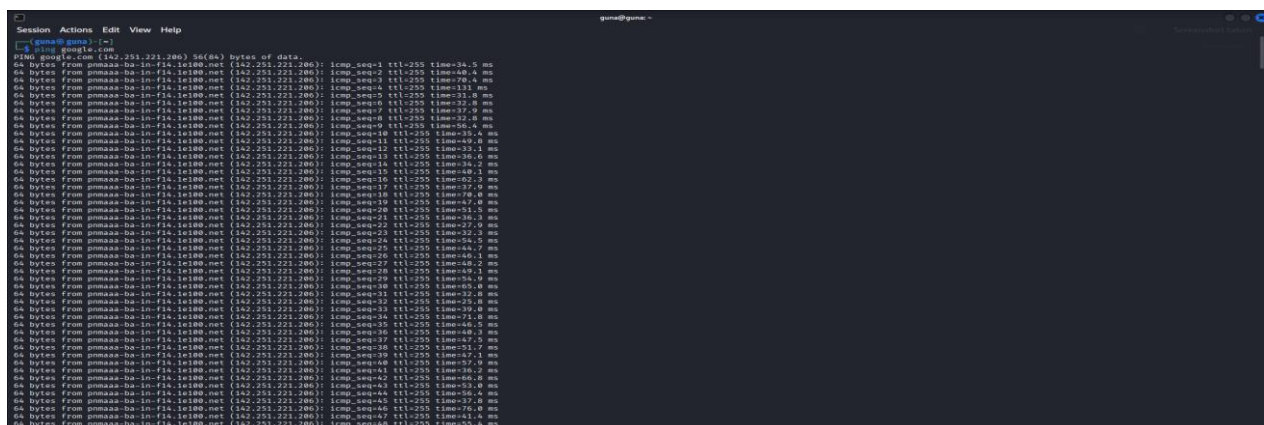
Tests connectivity

Helps identify network latency

Useful for basic troubleshooting in TCP/IP networks

Ping specific IP : ping 8.8.8.8

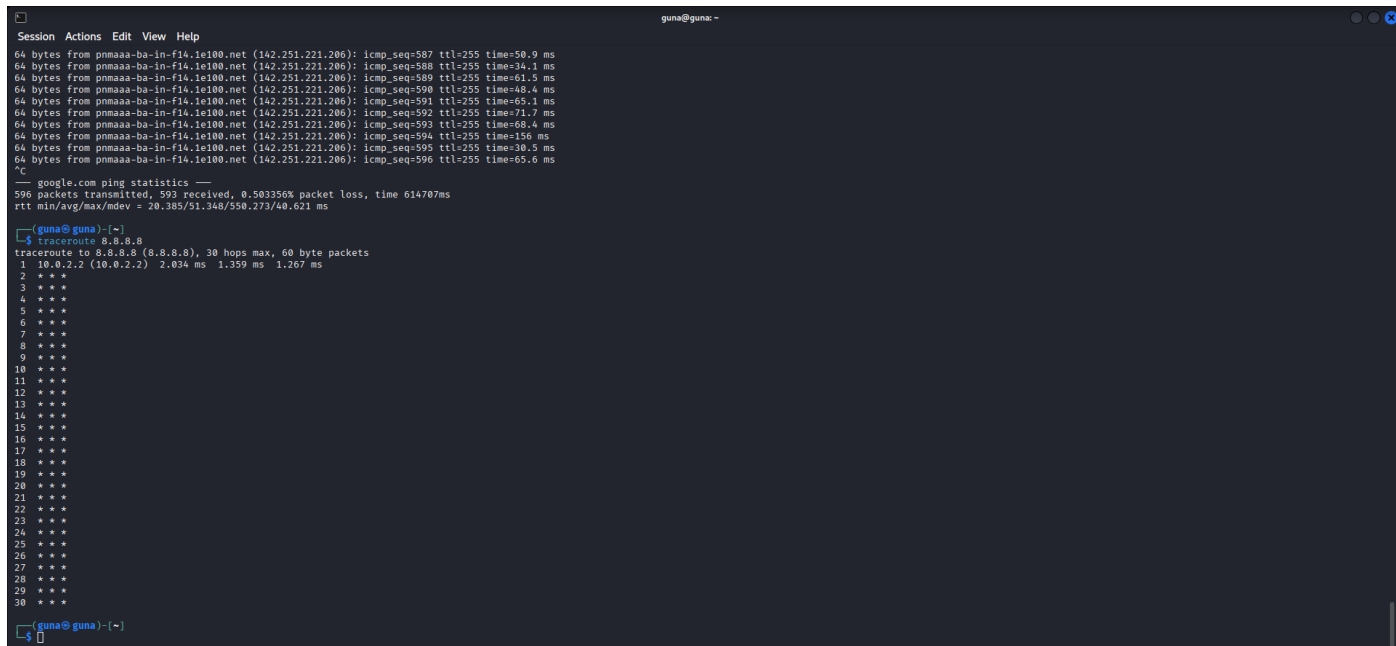
Send limited packets (very useful): ping -c 4 google.com



```
Session Actions Edit View Help
[~] (root@kali: ~)
root@kali:~# ping google.com
PING google.com (142.251.221.206) 56(84) bytes of data:
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=1 ttl=255 time=34.5 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=2 ttl=255 time=46.4 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=3 ttl=255 time=76.4 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=4 ttl=255 time=111 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=5 ttl=255 time=131.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=6 ttl=255 time=132.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=7 ttl=255 time=37.9 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=8 ttl=255 time=22.6 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=9 ttl=255 time=56.4 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=10 ttl=255 time=32.4 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=11 ttl=255 time=49.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=12 ttl=255 time=36.1 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=13 ttl=255 time=36.6 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=14 ttl=255 time=36.2 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=15 ttl=255 time=48.1 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=16 ttl=255 time=37.2 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=17 ttl=255 time=37.9 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=18 ttl=255 time=51.5 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=19 ttl=255 time=37.0 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=20 ttl=255 time=36.3 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=21 ttl=255 time=27.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=22 ttl=255 time=32.5 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=23 ttl=255 time=32.5 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=24 ttl=255 time=34.5 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=25 ttl=255 time=44.7 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=26 ttl=255 time=46.1 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=27 ttl=255 time=46.2 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=28 ttl=255 time=46.2 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=29 ttl=255 time=46.2 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=30 ttl=255 time=65.0 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=31 ttl=255 time=39.0 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=32 ttl=255 time=25.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=33 ttl=255 time=39.0 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=34 ttl=255 time=71.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=35 ttl=255 time=39.2 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=36 ttl=255 time=48.3 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=37 ttl=255 time=47.7 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=38 ttl=255 time=51.7 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=39 ttl=255 time=47.3 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=40 ttl=255 time=37.9 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=41 ttl=255 time=38.2 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=42 ttl=255 time=66.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=43 ttl=255 time=37.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=44 ttl=255 time=56.6 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=45 ttl=255 time=56.6 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=46 ttl=255 time=76.8 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=47 ttl=255 time=72.4 ms
64 bytes from pm15aa-ba-in-f14.1e100.net: icmp_seq=48 ttl=255 time=55.4 ms
```

2. Traceroute

- traceroute identifies the path packets take to reach a destination
 - Displays intermediate routers (hops)
 - Helps diagnose delays and routing problems
- Purpose of traceroute
 - Detect network bottlenecks
 - Identify failing hop
 - Understand routing path
 - What traceroute Actually Does
 - Sends packets with increasing TTL values
 - Each router decrements TTL
 - Router returns ICMP Time Exceeded

A screenshot of a terminal window titled 'guna@guna: ~'. The terminal shows the output of a traceroute command. At the top, it displays a series of ping statistics for a destination, showing 64 bytes from 'pnmaaa-ba-in-f14.1e100.net' with various TTL values and times. Below this, it shows 'google.com ping statistics' with 596 packets transmitted, 593 received, and 0.503356% packet loss. The main part of the output is the traceroute to 8.8.8.8, showing 30 hops. The first hop is 10.0.2.2 (10.0.2.2) with a 2.034 ms delay. The subsequent hops are marked with asterisks, indicating that the TTL has been exceeded. The terminal prompt is '\$'.

3. Netstat Command

netstat displays active network connections and listening ports

- Shows protocol statistics
 - Useful for network monitoring
- Purpose of netstat
 - View open connections
 - Identify listening services
 - Troubleshoot port issues

- t $\rightarrow$ TCP
- u $\rightarrow$ UDP
- l $\rightarrow$ Listening
- n $\rightarrow$ Numeric format

1. ss cmd :

Show active connections: `ss -an`

ss provides socket statistics

- ## Purpose of ss

- Example: ss -tuln

- Same use cases as netstat
- Preferred in newer systems

```

guna@guna ~
Session Actions Edit View Help
unix 3 [ ] STREAM CONNECTED 13749
unix 3 [ ] STREAM CONNECTED 16493
unix 3 [ ] STREAM CONNECTED 18885
unix 3 [ ] STREAM CONNECTED 13824
unix 3 [ ] STREAM CONNECTED 8999 /run/user/1000/bus
unix 3 [ ] STREAM CONNECTED 13735
unix 3 [ ] STREAM CONNECTED 13011 /run/dbus/system_bus_socket
unix 3 [ ] STREAM CONNECTED 12141 /run/user/1000/at-spi-bus_0
unix 2 [ACC] STREAM LISTENING 11008 0/tmp/.ICE-unix/1235
unix 3 [ ] STREAM LISTENING 7982 0/tmp/.X11-unix/X0
unix 3 [ ] STREAM CONNECTED 2893 0d0388411bd0883ac/bus/systemd-logind/system
unix 3 [ ] STREAM CONNECTED 11732 017523999516f789d/bus/systemd/bus-system
unix 3 [ ] STREAM CONNECTED 11992 00inter-applet-lock-user-guna
unix 3 [ ] STREAM CONNECTED 10178 0280bb68003c088bc/bus/systemd/bus-api-user
unix 3 [ ] STREAM CONNECTED 4000 0af46e3275fac627/bus/systemd/bus-api-system
unix 2 [ ] DGRAM 3622 02447791699181902352

(guna@guna)~$ ss -tlnl
State Recv-Q Send-Q Local Address:Port Peer Address:Port
tcp LISTEN 0 4096 127.0.0.1:36361 0.0.0.0:*

(guna@guna)~$ ss -sln
RTNETLINK answers: Invalid argument
State Recv-Q Send-Q Local Address:Port Peer Address:Port
nl UNCONN 0 0 0:0 *
nl UNCONN 0 0 0:1788 *
nl UNCONN 0 0 0:0 *
nl UNCONN 960 0 0:829 *
nl UNCONN 0 0 0:1788 *
nl UNCONN 0 0 0:679 *
nl UNCONN 960 0 4:0 *
nl UNCONN 4352 0 4:2184 *
nl UNCONN 0 0 6:0 *
nl UNCONN 0 0 6:829 *
nl UNCONN 0 0 7:0 *
nl UNCONN 0 0 9:1 *
nl UNCONN 0 0 9:0 *
nl UNCONN 0 0 10:0 *
nl UNCONN 0 0 11:0 *
nl UNCONN 0 0 12:829 *
nl UNCONN 0 0 12:1 *
nl UNCONN 0 0 12:0 *
nl UNCONN 0 0 15:-2033203276 *
nl UNCONN 0 0 15:-17431960 *
nl UNCONN 0 0 15:-930210202 *
nl UNCONN 0 0 15:1693 *
nl UNCONN 0 0 15:679 *
nl UNCONN 0 0 15:810 *
nl UNCONN 0 0 15:-1690689002 *
```

5. ip route Command (Check Gateway & Routes)

ip route displays routing table

- Shows how packets are forwarded

Purpose of ip route

- Identify default gateway
- Check routing paths
- Diagnose routing problems

Example: ip route

Important Output Entry

- default via <gateway IP>

Practical Use

- Troubleshoot Internet issues
- Detect wrong gateway
- Verify network configuration



```
guna@guna: ~  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
    valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host noprefixroute  
    valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
link/ether 00:00:27:00:10:c3 brd ff:ff:ff:ff:ff:ff  
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
    valid_lft 84663sec preferred_lft 84663sec  
inet6 fd17:625c:f037:2:a00:27ff:fe00:16cb/64 scope global temporary dynamic  
    valid_lft 86072sec preferred_lft 14072sec  
inet6 fd17:625c:f037:2:a00:27ff:fe00:16cb/64 scope global dynamic mngtmpaddr noprefixroute  
    valid_lft 86072sec preferred_lft 14072sec  
inet6 fe80::a00:27ff:fe00:16cb/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default  
link/ether 02:42:7b:a0:0f:a1 brd ff:ff:ff:ff:ff:ff  
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
    valid_lft forever preferred_lft forever  
  
guna@guna: ~  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
    valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host noprefixroute  
    valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
link/ether 00:00:27:00:10:c3 brd ff:ff:ff:ff:ff:ff  
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
    valid_lft 84658sec preferred_lft 84658sec  
inet6 fd17:625c:f037:2:a00:27ff:fe00:16cb/64 scope global temporary dynamic  
    valid_lft 86059sec preferred_lft 14059sec  
inet6 fd17:625c:f037:2:a00:27ff:fe00:16cb/64 scope global dynamic mngtmpaddr noprefixroute  
    valid_lft 86059sec preferred_lft 14059sec  
inet6 fe80::a00:27ff:fe00:16cb/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default  
link/ether 02:42:7b:a0:0f:a1 brd ff:ff:ff:ff:ff:ff  
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
    valid_lft forever preferred_lft forever  
  
guna@guna: ~  
$ ip route  
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100  
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100  
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown  
  
guna@guna: ~
```

6. telnet Command (IP Port Testing)

Test SSH port : telnet google.com 80

Test local machine port: telnet 127.0.0.1 22

- telnet can test connectivity to a specific IP and port
- Used as a simple port testing tool

Purpose of telnet

- Check if port is open
- Verify service availability
- Test firewall rules

Basic Syntax: telnet <IP address> <port>

Example: telnet 192.168.1.10 22

Result Interpretation

- Connected → Port open
- Connection refused → Port closed / blocked

Practical Use

- Service troubleshooting
- Firewall verification
- Port accessibility testing

```

guna@guna: ~
Session Actions Edit View Help
valid_lft 86072sec preferred_lft 14072sec
inet6 fe80::a00:27ff:fe00:16cb/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:76:a0:0f:a1 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever

(guna@guna)~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:00:16:c3 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
valid_lft 84650sec preferred_lft 84650sec
inet6 fd17:625c:f037:2::fa75:49d6:c982:92f9/64 scope global temporary dynamic
valid_lft 86089sec preferred_lft 14059sec
inet6 fd17:625c:f037:2::a00:27ff:fe00:16cb/64 scope global dynamic mngtmpaddr noprefixroute
valid_lft 86059sec preferred_lft 14059sec
inet6 fe80::a00:27ff:fe00:16cb/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:76:a0:0f:a1 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever

(guna@guna)~$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown

(guna@guna)~$ telnet google.com 80
Trying 142.251.222.142 ...
Connected to google.com.
Escape character is '^['.
Connection closed by foreign host.

(guna@guna)~$ telnet 127.0.0.1 22
Trying 127.0.0.1...
telnet: Unable to connect to remote host: Connection refused

(guna@guna)~$

```

- Check DNS resolution: nslookup google.com
- Check open ports on system: sudo ss -lntp
- Check default gateway quickly: ip route | grep default

```
guna@guna:~$ ip netns exec fd17:625c:f037:2:a00:27ff:fe00:16cb/64 scope global dynamic mngtmpaddr noprefixroute
valid_lft 88059sec preferred_lft 14059sec
inet6 fd00::a00:27ff:fe00:16cb/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:76:a0:0f:a1 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever

(guna@guna)~$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown

(guna@guna)~$ telnet google.com 80
Trying 142.251.222.142 ...
Connected to google.com.
Escape character is '^['.
Connection closed by foreign host.

(guna@guna)~$ telnet 127.0.0.1 22
Trying 127.0.0.1...
telnet: Unable to connect to remote host: Connection refused

(guna@guna)~$ nslookup google.com
Server: 10.178.120.105
Address: 10.178.120.105#53

Non-authoritative answer:
Name: google.com
Address: 142.251.43.206
Name: google.com
Address: 2404:6800:4007:821::200e

(guna@guna)~$ sudo ss -ltnp
[sudo] password for guna:
State      Recv-Q    Send-Q    Local Address:Port    Peer Address:Port    Process
LISTEN     0          4096      127.0.0.1:36361      0.0.0.0:*             users:((("containerd",pid=797,fd=9))

(guna@guna)~$ ip route | grep default
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100

(guna@guna)~$
```

File Viewing Commands in Linux

1. cat command

- Displays entire contents of a file
- Suitable for small files
- Outputs file directly to terminal

Example

```
cat filename
cat notes.txt
```

2. less command

- Displays file content page by page
- Allows scrolling up and down
- Best for large files

Example

```
less filename
less notes.txt
```

3. more command

- Displays file content one screen at a time
- Limited navigation compared to less
- Used for basic viewing

Example: more filename
more notes.txt

4. head command

- Displays beginning of a file
- Default shows first 10 lines
- Useful for previewing files

Example

head filename
head notes.txt

Custom lines: head -n 5 notes.txt

5. tail command

- Displays end of a file
- Default shows last 10 lines
- Commonly used for log inspection

Example

tail filename
tail notes.txt

Custom lines: tail -n 5 notes.txt

6. tail -f command

- Displays file updates in real time
- Used for monitoring logs
- Runs continuously until stopped

Example

tail -f filename
tail -f /var/log/syslog

7. grep command

- Searches for specific text inside file
- Displays matching lines only
- Very useful for troubleshooting

Example: grep "word" filename
grep error syslog

8. file command

- Identifies file type
- Shows whether file is text, binary, etc.

Example: file filename
file notes.txt

- cat → Show full file
- less → Scrollable viewer
- more → Basic page viewer
- head → First lines
- tail → Last lines
- tail -f → Live updates
- grep → Search text
- file → File type info

```

guna@guna: ~/linux_practice
Session Actions Edit View Help
Error: Network timeout

(guna@guna) ~/linux_practice
$ head notes.txt
Linux is powerful
Learning commands is important

(guna@guna) ~/linux_practice
$ tail logs.txt
Error: Disk full
Warning: CPU high
Error: Network timeout

(guna@guna) ~/linux_practice
$ tail -n 2 logs.txt
Warning: CPU high
Error: Network timeout

(guna@guna) ~/linux_practice
$ tail -f logs.txt
Error: Disk full
Warning: CPU high
Error: Network timeout

echo "Error: Unauthorized access" >> logs.txt
^C

(guna@guna) ~/linux_practice
$ grep "Error" logs.txt
Error: Disk full
Error: Network timeout

(guna@guna) ~/linux_practice
$ grep -i "error" logs.txt
Error: Disk full
Error: Network timeout

(guna@guna) ~/linux_practice
$ grep -c "Error" logs.txt
2

(guna@guna) ~/linux_practice
$ file notes.txt
notes.txt: ASCII text

(guna@guna) ~/linux_practice
$ file /bin/ls
/bin/ls: ELF 64-bit LSB pie executable, x86_64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0243f2a3ad64d635299a574bbc8ef951ddde9b21, for GNU/Linux 3.2.0, stripped

(guna@guna) ~/linux_practice

guna@guna: ~/linux_practice
Session Actions Edit View Help

(guna@guna) ~
$ mkdir linux_practice
cd linux_practice

(guna@guna) ~/linux_practice
$ touch notes.txt
touch logs.txt

(guna@guna) ~/linux_practice
$ echo "Linux is powerful" >> notes.txt
echo "Learning commands is important" >> notes.txt
echo "Error: Disk full" >> logs.txt
echo "Warning: CPU high" >> logs.txt
echo "Error: Network timeout" >> logs.txt

(guna@guna) ~/linux_practice
$ cat notes.txt
Linux is powerful
Learning commands is important

(guna@guna) ~/linux_practice
$ cat notes.txt logs.txt
Linux is powerful
Learning commands is important
Error: Disk full
Warning: CPU high
Error: Network timeout

(guna@guna) ~/linux_practice
$ less logs.txt

(guna@guna) ~/linux_practice
$ more logs.txt
Error: Disk full
Warning: CPU high
Error: Network timeout

(guna@guna) ~/linux_practice
$ head notes.txt
Linux is powerful
Learning commands is important

(guna@guna) ~/linux_practice
$ tail logs.txt
Error: Disk full
Warning: CPU high
Error: Network timeout

(guna@guna) ~/linux_practice
$ tail -n 2 logs.txt
Warning: CPU high

```

FILE & DIRECTORY COMMANDS

```
guna@guna: ~/linux_practice
Session Actions Edit View Help
Error: Network timeout

(guna@guna)-[~/linux_practice]
$ grep -c "Error" logs.txt
2

(guna@guna)-[~/linux_practice]
$ file notes.txt
notes.txt: ASCII text

(guna@guna)-[~/linux_practice]
$ file /bin/ls
/bin/ls: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0243f2a3ad64d635299a574bbc8ef951ddde9b21, for GNU/Linux 3.2.0, stripped

(guna@guna)-[~/linux_practice]
$ ls
ls -l          # detailed
ls -a          # hidden files
logs.txt notes.txt
total 8
-rw-rw-r-- 1 guna guna 58 Feb 26 17:49 logs.txt
-rw-rw-r-- 1 guna guna 49 Feb 26 17:49 notes.txt
. .. logs.txt notes.txt

(guna@guna)-[~/linux_practice]
$ cd ..
cd linux_practice

(guna@guna)-[~/linux_practice]
$ pwd
/home/guna/linux_practice

(guna@guna)-[~/linux_practice]
$ mkdir test_dir

(guna@guna)-[~/linux_practice]
$ cp notes.txt test_dir/

(guna@guna)-[~/linux_practice]
$ mv notes.txt renamed.txt
mv renamed.txt test_dir/

(guna@guna)-[~/linux_practice]
$ rm logs.txt
rm -r test_dir

(guna@guna)-[~/linux_practice]
$ touch newfile.txt

(guna@guna)-[~/linux_practice]
$
```

PERMISSIONS COMMANDS

```
guna@guna: ~/linux_practice
Session Actions Edit View Help
chmod: cannot access 'script.sh': No such file or directory

(guna@guna)-[~/linux_practice]
$ chmod 755 script.sh
chmod: cannot access 'script.sh': No such file or directory

(guna@guna)-[~/linux_practice]
$ sudo chown user:user filename
[sudo] password for guna:
chown: invalid user: 'user:user'

(guna@guna)-[~/linux_practice]
$ sudo chown guna:guna notes.txt
chown: cannot access 'notes.txt': No such file or directory

(guna@guna)-[~/linux_practice]
$ touch script.sh

(guna@guna)-[~/linux_practice]
$ ls
newfile.txt script.sh

(guna@guna)-[~/linux_practice]
$ nano script.sh

(guna@guna)-[~/linux_practice]
$ chmod +x script.sh

(guna@guna)-[~/linux_practice]
$ ls -l
total 4
-rw-rw-r-- 1 guna guna 0 Feb 26 18:04 newfile.txt
-rwxrwxr-x 1 guna guna 30 Feb 26 18:09 script.sh

(guna@guna)-[~/linux_practice]
$ -rw-r--r--
-rw-r--r--: command not found

(guna@guna)-[~/linux_practice]
$ ./script.sh
Hello Guna

(guna@guna)-[~/linux_practice]
$ sudo chown guna:guna script.sh

(guna@guna)-[~/linux_practice]
$ whoami
guna

(guna@guna)-[~/linux_practice]
$
```

1. chmod — Change File Permissions

Purpose:

Modifies access rights.

Why it is used:

- Making scripts executable
- Restricting sensitive files
- Hardening systems

Symbolic Mode:

```
chmod +x script.sh    # add execute
chmod -w file.txt     # remove write
```

Numeric Mode (Critical for Interviews):

```
chmod 755 script.sh
chmod 600 secret.txt
```

Meaning of Numbers:

- 7 → read + write + execute
- 6 → read + write
- 5 → read + execute
- 4 → read only

Security relevance:

Improper permissions → major vulnerability.

2. chown — Change File Ownership

Purpose:

Changes owner of file/directory.

Why it is used:

- Fixing permission errors
- Managing multi-user systems
- Web server/service setups

```
sudo chown user:user file.txt
```

3. ls -l — Permission Inspection

Purpose:

Displays permission structure.

Example:

```
-rwxr-xr-x
```

Meaning:

- Owner → rwx
- Group → r-x
- Others → r-x

Network Monitoring

```
Session Actions Edit View Help
(guna@guna)-[~]
$ ss -tulnp
Netid      State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port      Process
tcp        LISTEN     0            4096         127.0.0.1:41557          0.0.0.0:*

(guna@guna)-[~]
$ ss -nl
RTNETLINK answers: Invalid argument
Netid      State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port
nl         UNCONN     0            0            rtnl:kernel              *
nl         UNCONN     960          0            rtnl:dockerd/822         *
nl         UNCONN     0            0            rtnl:NetworkManager/665 *
nl         UNCONN     0            0            rtnl:xdg-desktop-por/1802 *
nl         UNCONN     0            0            rtnl:xdg-desktop-por/1802 *
nl         UNCONN     0            0            rtnl:NetworkManager/665 *
nl         UNCONN     960          0            tcpdiag:kernel           *
nl         UNCONN     4352         0            tcpdiag:ss/2279          *
nl         UNCONN     0            0            xfrm:dockerd/822         *
nl         UNCONN     0            0            xfrm:kernel              *
nl         UNCONN     0            0            selinux:kernel           *
nl         UNCONN     0            0            audit:systemd/1          *
nl         UNCONN     0            0            audit:kernel             *
nl         UNCONN     0            0            fiblookup:kernel         *
nl         UNCONN     0            0            connector:kernel         *
nl         UNCONN     0            0            nft:kernel               *
nl         UNCONN     0            0            nft:dockerd/822         *
nl         UNCONN     0            0            nft:systemd/1            *
nl         UNCONN     0            0            uevent:-1223009049       *
nl         UNCONN     0            0            uevent:Xorg/808          *
nl         UNCONN     0            0            uevent:gvfs-gphoto2-vo/1706 *
nl         UNCONN     0            0            uevent:gvfs-udisks2-vo/1686 *
nl         UNCONN     4800         0            uevent:Thunar/1442       *
nl         UNCONN     0            0            uevent:udisksd/1600      *
nl         UNCONN     0            0            uevent:wireplumber/1214  *
nl         UNCONN     0            0            uevent:systemd/1185      *
nl         UNCONN     0            0            uevent:systemd/1         *
nl         UNCONN     0            0            uevent:-1754802851       *
nl         UNCONN     0            0            uevent:gvfsd-metadata/1743 *
nl         UNCONN     4800         0            uevent:-1872023334       *
nl         UNCONN     0            0            uevent:-48799882         *
nl         UNCONN     0            0            uevent:-270760215        *
nl         UNCONN     0            0            uevent:-376118487        *
nl         UNCONN     0            0            uevent:-1747764394       *
nl         UNCONN     0            0            uevent:-204106500        *
nl         UNCONN     0            0            uevent:-46565394         *
nl         UNCONN     0            0            uevent:kernel            *
nl         UNCONN     0            0            uevent:-2138019679       *
nl         UNCONN     0            0            uevent:upowerd/1565      *
nl         UNCONN     0            0            uevent:-872219135        *
nl         UNCONN     0            0            uevent:ModemManager/684  *
nl         UNCONN     0            0            uevent:-1682877019       *
```

```
Session Actions Edit View Help
(guna@guna)-[~]
$ ss -tulnp | grep 80
u_str      LISTEN     0            4096         @/tmp/.X11-unix/X0 9919      * 0
u_str      UNCONN     0            0            @printer-applet-lock-user-guna 18570     * 0
u_dgr      UNCONN     0            0            @4072887113199411357 3613      * 0
icmp6      UNCONN     0            0            *:ipw6-icmp           *:*
tcp        LISTEN     0            4096         127.0.0.1:41557          0.0.0.0:*

(guna@guna)-[~]
$ ss -tulnp | grep 80
(guna@guna)-[~]
$ netstat -tulnp
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address      Foreign Address    State      PID/Program name
tcp        0      0 127.0.0.1:41557    0.0.0.0:*          LISTEN     -

(guna@guna)-[~]
$ sudo apt install net-tools
[sudo] password for guna:
net-tools is already the newest version (2.10-2).
net-tools set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0

(guna@guna)-[~]
$ netstat -tulnp
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address      Foreign Address    State      PID/Program name
tcp        0      0 127.0.0.1:41557    0.0.0.0:*          LISTEN     -

(guna@guna)-[~]
$ sudo ifconfig -i :22
(guna@guna)-[~]
$ sudo ifconfig -i :3000
(guna@guna)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:00:18:c0 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 84522sec preferred_lft 84522sec
```

```
Session Actions Edit View Help
(guna@guna)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:00:18:c3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86422sec preferred_lft 86422sec
    inet6 fd17:625c:f837:2:57bd:51c6:4b3:a9c0 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 8640sec preferred_lft 14240sec
    inet6 fe80::a0b:27ff:fe00:18c3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: dockero: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:00:54:00:15:00 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global dockero
        valid_lft forever preferred_lft forever

(guna@guna)-[~]
$ ping google.com
PING google.com (142.251.43.46) 56(84) bytes of data.
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=1 ttl=255 time=46.6 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=2 ttl=255 time=34.7 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=3 ttl=255 time=42.8 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=4 ttl=255 time=60.1 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=5 ttl=255 time=61.7 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=6 ttl=255 time=63.9 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=7 ttl=255 time=48.2 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=8 ttl=255 time=135 ms
64 bytes from pnm3aa-av-in-f14.1e100.net (142.251.43.46): icmp_seq=9 ttl=255 time=55.0 ms
^C
  google.com ping statistics ---
  9 packets transmitted, 0 received, 0% packet loss, time 8033ms
  rtt min/avg/max/mdev = 34.657/63.123/155.172/33.778 ms

(guna@guna)-[~]
$ curl http://example.com
^C

(guna@guna)-[~]
$ curl http://localhost:3000
curl: (7) Failed to connect to localhost port 3000 after 0 ms: Could not connect to server

(guna@guna)-[~]
$ wget http://example.com/file.txt
--2026-02-26 18:17:36--  http://example.com/file.txt
Resolving example.com (example.com)... 104.18.26.120, 104.18.27.120, 2606:4700:9702:8c7c:3376:502:ccc4:a209
Connecting to example.com (example.com)[104.18.26.120]:80... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-02-26 18:17:37 ERROR 404: Not Found.
```

```
Session Actions Edit View Help
(guna@guna)-[~]
$ curl http://example.com
^C

(guna@guna)-[~]
$ curl http://localhost:3000
curl: (7) Failed to connect to localhost port 3000 after 0 ms: Could not connect to server

(guna@guna)-[~]
$ wget http://example.com/file.txt
--2026-02-26 18:17:36--  http://example.com/file.txt
Resolving example.com (example.com)... 104.18.26.120, 104.18.27.120, 2606:4700:9702:8c7c:3376:502:ccc4:a209
Connecting to example.com (example.com)[104.18.26.120]:80... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-02-26 18:17:37 ERROR 404: Not Found.

(guna@guna)-[~]
$ watch ss -tulnp

(guna@guna)-[~]
$ sudo apt install iftop
Installing:
 iftop

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 38.9 kB
  Space needed: 95.2 kB / 10.1 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 iftop amd64 1.0~pre4-10 [38.9 kB]
Fetched 38.9 kB in 1s (40.4 kB/s)
Selecting previously unselected package iftop.
(Reading database... 559205 files and directories currently installed.)
Preparing to unpack .../iftop_1.0~pre4-10_amd64.deb...
Unpacking iftop (1.0~pre4-10)...
Setting up iftop (1.0~pre4-10)...
Processing triggers for kali-menu (2026.1.3)...
Processing triggers for man-db (2.13.1-1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.
```

```
Session Actions Edit View Help
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

(guna@guna)-[~]
$ sudo iftop
interface: eth0
IP address 1s: 10.0.2.15
IPv6 address 1s: fd17:625c:f837:2:57bd:51c6:4b3:a9c0
MAC address 1s: 08:00:27:00:18:c3

(guna@guna)-[~]
$ nmap 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 18:19 +0530
Nmap scan report for 192.168.1.1
Host is up (0.0013s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 5.41 seconds

(guna@guna)-[~]
$ ss -tulnp
sudo ss -l -t
State      LISTEN
Recv-Q    0
Send-Q    4096
Local Address:Port      Peer Address:Port      Process
COMMAND  PID USER FD      TYPE DEVICE SIZE/OFF  NODE NAME
NetworkMa 665 root 26u    IPv6 11337    0t0  UDP 10.0.2.15:bootpc->10.0.2.2:bootps
container 797 root  9u    IPv4 11361    0t0  TCP localhost:41557 (LISTEN)

(guna@guna)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=46.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=45.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=64.6 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=42.7 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=57.0 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=51.8 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=60.9 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=255 time=42.6 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=255 time=51.7 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=255 time=52.0 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=255 time=59.3 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=255 time=57.5 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=255 time=53.3 ms
^C
```

- Check open ports: ss -tulnp
- Identify suspicious service: sudo lsof -i :PORT
- Inspect process: ps -p PID

1. ss — Socket Statistics (Modern Standard)

Purpose:

Displays active connections & listening ports.

Why it is used:

- Identify open ports
- Detect exposed services
- Troubleshoot network issues
- Security monitoring

`ss -tulnp`

Shows:

- TCP / UDP services
- Listening ports
- Associated processes

Security relevance:

Unexpected listening ports → possible compromise.

2. netstat — Legacy Network Tool

Purpose:

Older alternative to ss.

`netstat -tulnp`

3. lsof -i — Process & Port Mapping

Purpose:

Shows which program uses a port.

Why it is used:

- Debugging server conflicts
- Malware analysis
- Incident response

`sudo lsof -i :22`

4. ip a — Interface & IP Information

Purpose:

Displays network interfaces.

Why it is used:

- Verify IP assignment

- Network troubleshooting
- Adapter diagnostics

ip a

5. ping — Connectivity Test

Purpose:

Checks reachability of host.

ping google.com

Used for diagnosing:

- Internet issues
- Packet loss
- DNS problems

6. curl — HTTP / API Testing

Purpose:

Fetch data from URLs.

Why it is used:

- Test web servers
- API debugging
- Service validation

curl http://localhost:3000

7. wget — File Download Utility

Purpose:

Downloads files via terminal.

wget http://example.com/file.txt

Common in automation & scripting.

8. watch — Repeated Monitoring

Purpose:

Runs command continuously.

watch ss -tulnp

Used for live diagnostics.

Stop:

CTRL + C

1. pwd - Print Working Directory

What it does: Shows you where the heck you are in the filesystem

Eg: Lost in the terminal? pwd is like asking "Mom, where am I?" and actually getting a straight answer.

2. ls - List Directory Contents

What it does: Shows you what files and folders are in your current location

Eg: ls is like opening your cupboard to see how messy it really is

Important variations: •

- ls -l : Long format (detailed info)
- ls -a : Shows hidden files (the stuff you didn't want to see)
 - ls -lh : Human-readable file sizes
 - ls -lt : Sort by modification time
 - ls -R : Recursive (shows everything, EVERYTHING!)

3. cd - Change Directory

What it does: Moves you from one folder to another

Eg: cd is your magical wardrobe to Narnia, except Narnia is just another folder with more files.

4. mkdir - Make Directory

What it does: Creates a new folder

Eg: mkdir is like Marie Kondo telling you to create a place for everything. Now you have no excuse for messy files!

Useful flags:

- mkdir -p path/to/nested/dirs : Creates parent directories as needed (lazy mode activated)

5. touch - Create Empty File

What it does: Creates a new empty file or updates timestamp of existing file

Eg: touch is like high-fiving a file into reality. No file? BOOM, now there is one!

6. cat - Concatenate and Display Files

What it does: Shows file contents in terminal

Think of it as: Opening a book and reading it super fast

Eg: Named after cats because it just dumps everything on your screen like a cat knocking stuff off a table.

7. cp - Copy Files/Directories

What it does: Makes a copy of files or folders

Think of it as: Ctrl+C and Ctrl+V

8. mv - Move/Rename Files

What it does: Moves files to new location OR renames them

9. rm - Remove Files/Directories

What it does: Deletes files FOREVER (no recycle bin!)

DANGER FLAGS:

- rm -r : Remove directories recursively
- rm -f : Force (no questions asked)

- `rm -rf` : MAXIMUM DANGER - removes everything without asking

10. `echo` - Display Text

What it does: Prints text to the terminal

11. `grep` - Search Text Patterns

What it does: Searches for text patterns in files

Useful flags:

- `grep -i` : Case insensitive
- `grep -r` : Recursive search
- `grep -n` : Show line numbers
- `grep -v` : Invert match (show lines that DON'T match)
- `grep -c` : Count matches

12. `find` - Search for Files

What it does: Finds files and directories based on criteria

Common patterns:

- `find . -name "filename"` : Find by name
- `find . -type f` : Find only files
- `find . -type d` : Find only directories
- `find . -size +10M` : Find files bigger than 10MB
- `find . -mtime -7` : Files modified in last 7 days

13. `chmod` - Change File Permissions

What it does: Changes who can read, write, or execute files

Permission basics:

- `r` = read (4)
- `w` = write (2)
- `x` = execute (1)
- Common: 755 (`rwxr-x`), 644 (`rw-r--r--`), 777 (`rwrxrwx` - danger!)

14. `chown` - Change File Owner

What it does: Changes the owner of files/directories

Syntax: `chown user: group filename`

15. `ps` - Process Status

What it does: Shows running processes

Useful variations:

- `ps aux` : Show all processes (the full monty)
- `ps aux | grep processname` : Find specific process

16. `kill` - Terminate Processes

What it does: Stops running processes

Signal levels:

- `kill PID` : Polite termination (`SIGTERM`)
- `kill -9 PID` : Force kill (`SIGKILL` - the nuclear option)

17. `top` / `htop` - Interactive Process Viewer

What it does: Real-time view of system processes

Navigation in `top`:

- Press 'q' to quit
- Press 'k' to kill a process

- Press 'M' to sort by memory
- Press 'P' to sort by CPU

18. df - Disk Free

What it does: Shows disk space usage

df -h : Human readable (GB instead of bytes)

19. du - Disk Usage

What it does: Shows how much space files/directories use

Useful flags:

- du -h : Human readable
- du -s : Summary only
- du -sh * : Summary of each item in current directory

20. tar - Archive Files

What it does: Creates and extracts compressed archives

Common patterns:

- tar -czf archive.tar.gz directory/ : Create compressed archive
- tar -xzf archive.tar.gz : Extract archive
- tar -tzf archive.tar.gz : List contents

Flags meaning:

- c = create, x = extract, t = list • z = gzip, v = verbose, f = file

```

Session Actions Edit View Help
(guna@guna) ~
$ pwd
/home/guna
(guna@guna) ~
$ ls
ls -l
ls -a
ls -lh
ls -lt
ls -R
192.168.1.10 ACS certl Desktop Documents Downloads guna guna.txt linux_practice Music nmap_vuln.txt output.txt Pictures Public Templates UDP.pcapng vee vee.txt Videos
total 5424
-rw-rw-r-- 1 guna guna 256 Dec 2 21:42 192.168.1.10
-rw-rw-r-- 1 guna guna 940 Dec 3 07:12 ACS
-rw-rw-r-- 1 guna guna 940 Dec 2 19:33 certl
drwxr-xr-x 3 guna guna 4096 Oct 6 21:09 Desktop
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Documents
drwxr-xr-x 2 guna guna 4096 Feb 21 14:45 Downloads
-rw-rw-r-- 1 guna guna 940 Dec 2 20:00 guna
-rw-rw-r-- 1 guna guna 33880 Dec 2 22:26 guna.txt
drwxr-xr-x 2 guna guna 4096 Feb 26 10:12 linux_practice
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Music
-rw-rw-r-- 1 guna guna 502 Dec 3 00:37 nmap_vuln.txt
-rw-rw-r-- 1 guna guna 254 Dec 2 21:40 output.txt
drwxr-xr-x 2 guna guna 4096 Feb 24 10:05 Pictures
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Public
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Templates
-rw-rw-r-- 1 guna guna 5435748 Sep 29 21:40 UDP.pcapng
-rw-rw-r-- 1 guna guna 940 Dec 2 20:09 vee
-rw-rw-r-- 1 guna guna 11291 Dec 3 00:08 vee.txt
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Videos
. .sharpSuite Documents gufs Music .ssh .vboxclient-display-svga-x11-tty7-service.pid .vboxclient-vmvga-session-tty7-service.pid .xsession-errors
.cache Downloads .ICAuthority nmap_vuln.txt .sudo_as_admin_successful .vboxclient-draganddrop-tty7-control.pid .vboxclient-hostversion-tty7-control.pid .xsession-errors-old
192.168.1.10 certl .face .java output.txt Templates .vboxclient-draganddrop-tty7-service.pid vee.txt
ACS .config .face.icon linux_practice Pictures UDP.pcapng .vboxclient-hostversion-tty7-control.pid Videos
.bash_logout .dbus .gnupg .local .pkli .vboxclient-clipboard-tty7-control.pid .vboxclient-seamless-tty7-control.pid .viminfo
.bashrc Desktop guna mozilla .profile .vboxclient-clipboard-tty7-service.pid .vboxclient-seamless-tty7-service.pid .wget-hsts
.bashrc-original .dircr guna.txt .ssh .vboxclient-display-svga-x11-tty7-control.pid .vboxclient-vmvga-session-tty7-control.pid .zsh_history
total 5.3M
-rw-rw-r-- 1 guna guna 256 Dec 2 21:42 192.168.1.10
-rw-rw-r-- 1 guna guna 940 Dec 3 07:12 ACS
-rw-rw-r-- 1 guna guna 940 Dec 2 19:33 certl
drwxr-xr-x 3 guna guna 4096 Oct 6 21:09 Desktop
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Documents
drwxr-xr-x 2 guna guna 4096 Feb 21 14:45 Downloads
-rw-rw-r-- 1 guna guna 940 Dec 2 20:00 guna
-rw-rw-r-- 1 guna guna 34K Dec 2 22:26 guna.txt
drwxr-xr-x 2 guna guna 4096 Feb 26 10:12 linux_practice
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Music
-rw-rw-r-- 1 guna guna 502 Dec 3 00:37 nmap_vuln.txt

```

```

Session Actions Edit View Help
drwxr-xr-x 2 guna guna 4096 Feb 26 10:12 linux_practice
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Music
-rw-rw-r-- 1 guna guna 502 Dec 3 00:37 nmap_vuln.txt
-rw-rw-r-- 1 guna guna 254 Dec 2 21:40 output.txt
drwxr-xr-x 2 guna guna 4096 Feb 24 10:05 Pictures
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Public
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Templates
-rw-rw-r-- 1 guna guna 528 Sep 29 21:40 UDP.pcapng
-rw-rw-r-- 1 guna guna 940 Dec 2 20:09 vee
-rw-rw-r-- 1 guna guna 12K Dec 3 00:08 vee.txt
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Videos
total 5424
drwxr-xr-x 2 guna guna 4096 Feb 26 10:12 linux_practice
drwxr-xr-x 2 guna guna 4096 Feb 24 10:05 Pictures
drwxr-xr-x 2 guna guna 4096 Feb 21 14:45 Downloads
-rw-rw-r-- 1 guna guna 940 Dec 3 07:12 ACS
-rw-rw-r-- 1 guna guna 502 Dec 3 00:37 nmap_vuln.txt
-rw-rw-r-- 1 guna guna 256 Dec 2 21:42 192.168.1.10
-rw-rw-r-- 1 guna guna 33880 Dec 2 22:26 guna.txt
-rw-rw-r-- 1 guna guna 254 Dec 2 21:40 output.txt
-rw-rw-r-- 1 guna guna 940 Dec 2 20:09 vee
-rw-rw-r-- 1 guna guna 940 Dec 2 20:00 guna
drwxr-xr-x 3 guna guna 4096 Oct 6 21:09 Desktop
-rw-rw-r-- 1 guna guna 5435748 Sep 29 21:40 UDP.pcapng
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Documents
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Music
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Public
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Templates
drwxr-xr-x 2 guna guna 4096 Sep 22 14:57 Videos
.
192.168.1.10 ACS certl Desktop Documents Downloads guna guna.txt linux_practice Music nmap_vuln.txt output.txt Pictures Public Templates UDP.pcapng vee vee.txt Videos
./Desktop:
pentestlab
./Desktop/pentestlab:
install_docker_kali_x64.sh pentestlab.sh README.md
./Documents:
./Downloads:
'Lens Visualization [2026-02-21T14.44.32.315+05.30].csv' report-50e2d39b-0cde-4f07-b662-632a762236c4.pdf report-f731d111-99c7-4595-905b-7925fd266cd-1.pdf target-a4b2865e-2dff-4e30-a700-0841a0e94297.xml
'vessius-10-11:0-debinfo_endofa(1).deb' report-50e2d39b-0cde-4f07-b662-632a762236c4.xml report-f731d111-99c7-4595-905b-7925fd266cd.pdf targets-20251007.xml
'vessius-10-11:0-debinfo_endofa_end' report-78c067ca-bedd-ac5d-baff-f020bacc3083.pdf sam.pdf
./linux_practice:
newfile.txt script.sh
./Music:

```

```
guna@guna: ~  
Session Actions Edit View Help  
newfile.txt script.sh  
./Music:  
./Pictures:  
'exp 1.png' 'exp 2.png' 'exp4.png' 'exp 5.3.png' 'exp 6.png' 'exp8.2.png' 'exper1 4.2.png' 'experri 8.png' Screenshot_2026-02-24_16_05_18.png  
'exp 2.1.png' 'exp3.png' 'exp 5.1.png' 'exp 5.png' 'exp 7.png' 'exp8.png' 'exper1 4.3.png' Screenshot_2025-12-02_22_36_49.png  
'exp2.2.png' 'exp4.1.png' 'exp 5.2.png' 'exp 6.1.png' 'exp8.1.png' 'exper1 4.1.png' 'exper1 4.png' Screenshot_2026-02-20_21_33_18.png  
./Public:  
./Templates:  
./Videos:  
--(guna@guna)-[~]  
--$ cd docs  
cd: no such file or directory: docs  
--(guna@guna)-[~]  
--$ pwd  
/home/guna  
--(guna@guna)-[~]  
--$ cd ..  
cd..: command not found  
--(guna@guna)-[~]  
--$ mkdir test_dir  
--(guna@guna)-[~]  
--$ mkdir -p parent/child/grandchild  
--(guna@guna)-[~]  
--$ touch newfile.txt  
--(guna@guna)-[~]  
--$ ls -l newfile.txt  
-rw-rw-r-- 1 guna guna 0 Feb 26 19:35 newfile.txt  
--(guna@guna)-[~]  
--$ cat file1.txt  
cat: file1.txt: No such file or directory  
--(guna@guna)-[~]  
--$ cd file1.txt  
cd: no such file or directory: file1.txt  
--(guna@guna)-[~]  
--$ mkdir file1.txt
```

```
guna@guna: ~  
Session Actions Edit View Help  
--$ cd file1.txt  
cd: no such file or directory: file1.txt  
--(guna@guna)-[~]  
--$ mkdir file1.txt  
--(guna@guna)-[~]  
--$ cat file1.txt  
cat: file1.txt: Is a directory  
--(guna@guna)-[~]  
--$ cat notes.log  
cat: notes.log: No such file or directory  
--(guna@guna)-[~]  
--$ cat file1.txt file2.txt  
cat: file1.txt: Is a directory  
cat: file2.txt: No such file or directory  
--(guna@guna)-[~]  
--$  
--(guna@guna)-[~]  
--$ mkdir file2.txt  
--(guna@guna)-[~]  
--$ cat file2.txt  
cat: file2.txt: Is a directory  
--(guna@guna)-[~]  
--$ cat file1.txt file2.txt  
cat: file1.txt: Is a directory  
cat: file2.txt: Is a directory  
--(guna@guna)-[~]  
--$ cp file1.txt backup/  
cp: -r not specified; omitting directory 'file1.txt'  
--(guna@guna)-[~]  
--$ cp -r docs backup/  
cp: cannot stat 'docs': No such file or directory  
--(guna@guna)-[~]  
--$ mv file2.txt renamed.txt  
--(guna@guna)-[~]  
--$ mv renamed.txt docs/  
--(guna@guna)-[~]  
--$ rm newfile.txt
```

```
guna@guna: ~  
Session Actions Edit View Help  
--$ cat file1.txt file2.txt  
cat: file1.txt: Is a directory  
cat: file2.txt: Is a directory  
--(guna@guna)-[~]  
--$ cp file1.txt backup/  
cp: -r not specified; omitting directory 'file1.txt'  
--(guna@guna)-[~]  
--$ cp -r docs backup/  
cp: cannot stat 'docs': No such file or directory  
--(guna@guna)-[~]  
--$ mv file2.txt renamed.txt  
--(guna@guna)-[~]  
--$ mv renamed.txt docs/  
--(guna@guna)-[~]  
--$ rm newfile.txt  
--(guna@guna)-[~]  
--$ rm -r test_dir  
--(guna@guna)-[~]  
--$ echo "Hello Guna"  
Hello Guna  
--(guna@guna)-[~]  
--$ echo "New log entry" >> notes.log  
--(guna@guna)-[~]  
--$ grep "Error" notes.log  
--(guna@guna)-[~]  
--$ grep -l warning notes.log  
--(guna@guna)-[~]  
--$ grep -n Error notes.log  
--(guna@guna)-[~]  
--$ grep -c Error notes.log  
0  
--(guna@guna)-[~]  
--$ find . -name "file1.txt"  
find . -type f  
find . -type d  
find . -size +1M  
find . -mtime -1  
./file1.txt
```

```
guna@guna: ~
Session Actions Edit View Help

(guna@guna)-[~]
$ ls -l file1.txt
total 0

(guna@guna)-[~]
$ chmod 777 file1.txt

(guna@guna)-[~]
$ chmod 644 file1.txt

(guna@guna)-[~]
$ chmod +x file1.txt

(guna@guna)-[~]
$ whoami
Command 'whoami' not found, did you mean:
Command 'whoami' from deb coreutils
Command 'who' from deb mh
Command 'whom' from deb mailutils-mh
Try: sudo apt install <deb name>

(guna@guna)-[~]
$ whoami
guna

(guna@guna)-[~]
$ sudo chown guna:guna file1.txt
[sudo] password for guna:

(guna@guna)-[~]
$ ps
  PID TTY          TIME CMD
 2924 pts/1    00:00:01 zsh
 3254 pts/1    00:00:00 ps

(guna@guna)-[~]
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1   0.0  0.1 24872 15700 ?        Ss   17:45   0:01 /sbin/init splash
root           2   0.0  0.0      0   0 ?        S    17:45   0:00 [kthreadd]
root           3   0.0  0.0      0   0 ?        S    17:45   0:00 [pool_workqueue_release]
root           4   0.0  0.0      0   0 ?        Ic   17:45   0:00 [kworker/R-rcu_gp]
root           5   0.0  0.0      0   0 ?        Ic   17:45   0:00 [kworker/R-sync_wq]
root           6   0.0  0.0      0   0 ?        Ic   17:45   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7   0.0  0.0      0   0 ?        Ic   17:45   0:00 [kworker/R-slub_flushwq]
root           8   0.0  0.0      0   0 ?        Ic   17:45   0:00 [kworker/R-netns]
root          10   0.0  0.0      0   0 ?        Ic   17:45   0:00 [kworker/0:0H-events_highpri]
root          11   0.0  0.0      0   0 ?        I   17:45   0:01 [kworker/0:1-events]
root          12   0.0  0.0      0   0 ?        I   17:45   0:00 [kworker/u32:0]
root          13   0.0  0.0      0   0 ?        Ic   17:45   0:00 [kworker/R-mm_percpu_wq]
```

```
guna@guna: ~
Session Actions Edit View Help

root          3200  0.0  0.0      0   0 ?        I   19:41   0:00 [kworker/u33:1-events_unbound]
root          3250  0.0  0.0      0   0 ?        I   19:44   0:00 [kworker/5:0-ata_sff]
guna          3266  0.0  0.0    9808 4408 pts/1    R+   19:45   0:00 ps aux

(guna@guna)-[~]
$ ps aux | grep bash
guna          3274  0.0  0.0    6748 2384 pts/1    S+   19:45   0:00 grep --color=auto bash

(guna@guna)-[~]
$ sleep 100
^C

(guna@guna)-[~]
$ sleep 1

(guna@guna)-[~]
$ sleep 10

(guna@guna)-[~]
$ ps aux | grep sleep
guna          3293  0.0  0.0    6748 2388 pts/1    S+   19:48   0:00 grep --color=auto sleep

(guna@guna)-[~]
$ kill PID
kill: illegal pid: PID

(guna@guna)-[~]
$ kill -9 PID
kill: illegal pid: PID

(guna@guna)-[~]
$ top
top - 19:51:28 up 2:06, 1 user, load average: 0.02, 0.04, 0.00
Tasks: 246 total, 1 running, 245 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.1 us,  0.2 sy,  0.0 ni, 99.7 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  8241.4 total,  5047.1 free,  1115.9 used,  1815.2 buff/cache
MiB Swap:  2135.0 total,  2135.0 free,  0.0 used,  7125.6 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 680 root        20   0   87408 219296 126536 S 11.0  2.6   2:09cc Xorg
2027 guna       20   0   807036 70840 55048 S  0.9  0.8   0:06.00 qterminal
797 root       20   0 1621580 49808 36652 S  0.3  0.6   0:23.45 containerd
1312 guna       20   0 215864 3360 2900 S  0.3  0.0   0:12.25 VBoxClient
1320 guna       20   0 215580 3284 2960 S  0.2  0.0   0:49.20 VBoxClient
1398 guna       20   0 671740 132168 91612 S  0.3  1.6   0:32.75 xfwm4
1459 guna       20   0 339644 91792 21592 S  0.3  1.1   0:28.08 wrapper-2.0
1565 root       20   0 319320 11368 9116 S  0.3  0.1   0:09.11 upowerd
2787 root       20   0 0 0 0 I  0.3  0.0   0:05:57 kworker/5:2H-events_freezable_pwr_efficient
1 root       20   0 24872 15700 11296 S  0.0  0.2   0:01.84 systemd
2 root       20   0 0 0 0 S  0.0  0.0   0:00.20 kthreadd
3 root       20   0 0 0 0 S  0.0  0.0   0:00.00 pool_workqueue_release
```

```
guna@guna: ~
Session Actions Edit View Help

23 root       rt  0  0  0  0 S  0.0  0.0   0:00.81 migration/1
24 root       20  0  0  0  0 S  0.0  0.0   0:00.81 ksoftirqd/1
27 root       20  0  0  0  0 S  0.0  0.0   0:00.00 cpuhp/2
28 root      -51  0  0  0  0 S  0.0  0.0   0:00.00 idle_inject/2
29 root       rt  0  0  0  0 S  0.0  0.0   0:00.86 migration/2
30 root       20  0  0  0  0 S  0.0  0.0   0:00.00 ksoftirqd/2
32 root      -20  0  0  0  0 I  0.0  0.0   0:00.00 kworker/2:0H-events_highpri
33 root       20  0  0  0  0 S  0.0  0.0   0:00.00 cpuhp/3
34 root      -51  0  0  0  0 S  0.0  0.0   0:00.00 idle_inject/3
35 root       rt  0  0  0  0 S  0.0  0.0   0:00.88 migration/3
36 root       20  0  0  0  0 S  0.0  0.0   0:00.00 ksoftirqd/3
38 root      -20  0  0  0  0 I  0.0  0.0   0:00.00 kworker/3:0H-events_highpri
39 root       20  0  0  0  0 S  0.0  0.0   0:00.00 cpuhp/4
40 root      -51  0  0  0  0 S  0.0  0.0   0:00.00 idle_inject/4

(guna@guna)-[~]
$ htop
Command 'htop' not found, but can be installed with:
sudo apt install htop
Do you want to install it? (N/y)
sudo apt install htop
Installing:
 htop

Suggested packages:
 strace

Summary:
 Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
Download size: 171 kB
Space needed: 434 kB / 10.1 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 htop amd64 3.4.1-5 [171 kB]
Fetched 171 kB in 16s (11.0 kB/s)
Selecting previously unselected package htop.
(Reading database... 559212 files and directories currently installed.)
Preparing to unpack ./htop_3.4.1-5_amd64.deb...
Unpacking htop (3.4.1-5)...
Setting up htop (3.4.1-5)...
Processing triggers for kali-menu (2026.1.3)...
Processing triggers for desktop-file-utils (0.28-1)...
Processing triggers for hicolor-icon-theme (0.18-2)...
Processing triggers for man-db (2.13.1-1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.
```

```
guna@guna: ~
Session Actions Edit View Help
No VM guests are running outdated hypervisor (qemu) binaries on this host.

(guna@guna)~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
udev            4113936      0    4113936   0% /dev
tmpfs           843924    1036    842888   1% /run
/dev/sda1      38818336 26939488  9874420  74% /
tmpfs          4219620      4    4219616   1% /dev/shm
tmpfs          4219620 22848    4196772   1% /tmp
none           1024      0    1024    0% /run/credentials/getty@tty1.service
tmpfs          843924    108    843816   1% /run/user/1000
none           1024      0    1024    0% /run/credentials/systemd-journald.service

(guna@guna)~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            4.0G      0  4.0G   0% /dev
tmpfs           825M    1.1M  824M   1% /run
/dev/sda1       38G    26G  9.5G  74% /
tmpfs           4.1G  4.0K  4.1G   1% /dev/shm
tmpfs           4.1G   23M  4.1G   1% /tmp
none           1.0M      0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           825M   108K  825M   1% /run/user/1000
none           1.0M      0  1.0M   0% /run/credentials/systemd-journald.service

(guna@guna)~$ du -h
8.0K  ./java/.userPrefs/burp/community/detached-frames
16K  ./java/.userPrefs/burp/community
1.5M  ./java/.userPrefs/burp
1.5M  ./java/.userPrefs/burp
32K  ./java/fonts/21.0.9-ea
36K  ./java/fonts
1.6M  ./java
4.0K  ./Public
4.0K  ./parent/child/grandchild
8.0K  ./parent/child
12K  ./parent
4.0K  ./gnupg/private-keys-v1.d
8.0K  ./gnupg
3.4M  ./Pictures
4.0K  ./file1.txt
4.0K  ./Videos
4.0K  ./ssh/agent
8.0K  ./ssh
8.0K  ./linux_practice
4.0K  ./Music
4.0K  ./Documents
4.0K  ./BurpSuite/ConfigLibrary
4.0K  ./BurpSuite/bambdas
```

```
guna@guna: ~
Session Actions Edit View Help
72K  ./local/share/gvfs-metadata
8.0K  ./local/share/nautilus/scripts
12K  ./local/share/nautilus
4.0K  ./local/share/lxc
4.0K  ./local/share/nano
136K  ./local/share
12K  ./local/state/wireplumber
28K  ./local/state
164K  ./local
72K  ./pkg/nsddb
78K  ./pkg
1.3G  .

(guna@guna)~$ du -sh *
4.0K  192.168.1.10
4.0K  ACS
4.0K  certi
264K  Desktop
4.0K  docs
4.0K  Documents
130M  Downloads
4.0K  file1.txt
4.0K  guna
36K  guna.txt
8.0K  linux_practice
4.0K  Music
4.0K  mmap_vuln.txt
4.0K  notes.log
4.0K  output.txt
12K  parent
3.4M  Pictures
4.0K  Public
4.0K  Templates
5.2M  UDP.pcapng
4.0K  vee
12K  vee.txt
4.0K  Videos

(guna@guna)~$ tar -czf backup.tar.gz docs/

(guna@guna)~$ tar -tzf backup.tar.gz
docs/

(guna@guna)~$ tar -xzf backup.tar.gz

(guna@guna)~$
```